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Summary

Associate Professor of Biomedical Engineering by appointment with doctoral training in Pathology and an internationally visible research record, including publications in *The Lancet* family, *Scientific Reports*, *Nucleic Acids Research* and *Nature Medicine*. Leads an independent research program in translational and applied biomedical engineering informed by molecular pathology, with emphasis on infectious diseases, diagnostics, and clinically relevant biomedical systems, supported by competitive principal investigator-led funding. Contributes to large-scale international collaborations, including the Global Health Metrics & Evaluation and Global Burden of Diseases studies, which complement and contextualize the core independent research agenda. Has held sustained senior academic leadership roles, including dean-level appointments and chairmanship of a JFDA-approved Institutional Review Board. Recent work includes the identification and registration of globally novel *Streptococcus pneumoniae* sequence types in international reference databases, e.g. pubMLST, reflecting continued engagement in pathogen characterization and translational microbiology.

Education

Fellowship in Responsible Conduct of Research (RCR) | One-Year | University of California San Diego (UCSD), funded by John E. Fogarty International Center for Advanced Study in the Health Sciences at the U.S. National Institutes of Health ([NIH](#)), USA 2018

- Completed coursework: Drug Clinical Trials, Research Methods and Ethics.
- Epidemiology Methods and Ethics and Summer Practicum.

Ph.D. in Pathology | Faculty of Medicine | The University of New South Wales (UNSW), Australia 2015

- Title of Ph.D. Thesis: “Mechanism of DNA Binding of an Unusual Bacterial Transcription Regulator”

M.Sc. in Biotechnology | Yarmouk University (YU), Jordan 2009

- Title of M.Sc. Thesis: “Generation and Characterization of Glu 29 Arg Mutant *Drosophila* Dihydrofolate Reductase and Analysis of Methotrexate Resistance”

B.Sc. in Biotechnology & Genetic Engineering | Jordan University of Science & Technology (JUST), Jordan 2003

- Focus: molecular biology techniques, genetic analysis, and bioprocess fundamentals.

Academic Appointments

Associate Professor | School of Applied Medical Sciences, German Jordanian University 2020-Present

- M.Sc. Supervision & Examination.
- Served as Acting Vice Dean (2020-2021).

- Chair, Institutional Review Board (JFDA-approved) and Assistant to Dean of Scientific Research for Research Ethics (2025-present).

Assistant Professor | School of Applied Medical Sciences, German Jordanian University 2016-2020

- Served on multiple departmental and university committees.
- Served as Acting Vice Dean (2019-2020), Acting Head of Biomedical Engineering Department (2017-2020), and Assistant Dean for Accreditation & Quality Assurance (2017-2018).
- Contributed to the establishment of quality assurance frameworks and supported the school's accreditation efforts.

Assistant Professor | School of Pharmacy, Al-Zaytoonah University of Jordan 2015-2016

- Taught courses in pharmaceutical microbiology, biotechnology, and Human Physiology.
- Engaged in curriculum development and student advising.

Academic Facilitator | School of Medical Sciences, University of New South Wales, Australia 2014-2015

- Facilitated problem-based learning tutorials and small-group discussions for medical students.
- Supported course delivery in pathology and biomedical sciences, contributing to student assessment and feedback.

Professional & Industrial Experience

Medical Biotechnologist | Central Laboratories Directorate, Ministry of Health, Jordan October 2004-June 2006

- Performed diagnostic testing and quality control in a GLP-accredited environment.
- Gained expertise in laboratory safety protocols, standard operating procedures, and documentation.
- Supported public health initiatives through routine and specialized clinical analyses.

Medical Representative | IPSEN, Petra Drug Store, Jordan July 2006-September 2008

- Promoted pharmaceutical products to healthcare professionals, building strong client relationships.
- Delivered product presentations and provided clinical information to physicians and pharmacists.
- Monitored market trends and competitor activity to support sales strategies.

Product Manager | Arcomex, Medical Diagnostics Manufacturing, Jordan September 2008-August 2010

- Oversaw product lifecycle for diagnostic medical devices in a GMP-compliant manufacturing setting.
- Coordinated cross-functional teams including R&D, production, and quality assurance.
- Managed product launches, training, and post-market surveillance.

Medical Technologist | King Hussein Cancer Center (KHCC), Jordan August 2010-August 2011

- Conducted clinical laboratory testing in a GLP-compliant and USA-CAP-accredited oncology center.
- Operated advanced diagnostic equipment and ensured accuracy of test results.
- Collaborated with clinical teams to support patient care and research protocols.

Research Profile

Genomics Laboratory Leadership

- **Founder & Principal Investigator (PI)** | GJU Genomics Lab | German Jordanian University | 2023 – Present
 - Established and lead a dedicated genomics research facility at GJU with core technologies: Next-Generation Sequencing (NGS), real-time PCR (qPCR), and associated molecular biology platforms. Oversee lab operations, equipment acquisition, protocol development, and biosafety compliance and supervise and mentor graduate and undergraduate student researchers in genomics projects.
 - Drive research in infectious diseases, molecular pathology, and biomarker discovery using high-throughput genomic approaches.
 - Collaborate with local and international partners on translational genomics studies.

Full peer-reviewed publication list available on:

- [Scopus Author ID](#): 57202970794
- [Web of Science Researcher ID](#): JBJ-7585-2023
- [ORCID](#): 0000-0002-3772-5617
- [Google Scholar](#): Walid A. Al-Zyoud
- [ResearchGate](#): Walid A. Al-Zyoud

Research Interests

My research focuses on molecular pathology, biotechnology, and biomedical applications with a focus on developing laboratory, computational, and bioimaging methods for disease identification. A one-year clinical research fellowship at UC San Diego provided formal training in drug clinical trial methods, research ethics, and epidemiological study design, strengthening an ongoing commitment to global and translational research in infectious, respiratory, and viral diseases (e.g., COVID-19).

Research Awarded Funds

- **Received** Research Project Grant “Human Oral Microbiota Study in Tobacco and Hookah Young Jordanian Smokers as Dentists Prediction Indicator to Oral Diseases or Cancer” (SAMS28/2016): 29,692 JOD ≈ **\$41,890 USD** from GJU.
- **Received** Research Project Grant “Monitoring of coral fish parasites and diseases, an effective tool for coastal environmental health assessment” (2017) 29,100 JOD ≈ **\$41,052 USD** from the national Scientific Research Fund (SRF), Ministry of Higher Education- Higher Council of Science & Technology, Jordan
- **Received** Graduation project fund “Salivary microbiome of COVID-19 patients” (SAMS/2022): 997 JOD ≈ **\$1,408** from Deanship of Scientific research at GJU.
- **Received** Graduation project fund “Effect of COVID-19 on breast milk” (SAMS/2022): 997 JOD ≈ **\$1,408** from Deanship of Scientific research at GJU.
- **Received** Research Project Grant “Genomics studies to design coronavirus vaccine design for Jordanian population” (SAMS01/2022): 67,156 JOD ≈ **\$94,853 USD** from GJU.
- **Received** a Matching Grant “Revolutionizing diagnostics by suppressing coffee-ring effect in low antigen dot-blotting on electrospun PVDF membranes” (2025): 100,000

Malaysian Ringgit (MYR) $\approx$ **\$24,650 USD** co-funded by Universiti Teknologi MARA (UiTM), Malaysia

- **Received** for university capacity building “Selection GJU as Core University Partner for the Higher Education for Innovation and Growth Activity (HEIG), for the second cycle, focusing on the Medical Manufacturing Sector” (2025): **\$100,000 USD** from IREX-USAID, USA
- **Received** a project grant “Advancing Access to 3D Bioprinting for Medical Biomaterials”: 4,300 JOD $\approx$ **\$6,076** from King Abdullah II Fund for Development (KAJD) for university research and creativity projects in cooperation with Jordan Design and Development Bureau (JODDB), Jordan
- **Submitted (under final assessment** No. 397/7/5/1 dated 12/03/2026) "Production of whey protein allocated through fine fermentation of Jordanian tasted chocolate": 40,000 JOD $\approx$ **\$56,521**, submitted to the food security program (third round) by the Higher Council for Science and Technology (HCST), Jordan

Grants Submitted, shortlisted but Not Awarded:

- “RNA based anti COVID19 vaccine design for Jordan”, MPH/1/18/ (2022): 29,000 JOD $\approx$ **40,000 USD**, submitted to The Ministry of Higher Education and Scientific Research, the Scientific Research Support Fund (SRF), Jordan
- “URCHIN-BASE: Understanding Reasons for Collapse in Habitat and Infection in Sea Urchins - Baseline Analysis Study Evaluation” 232/9/1(2024): 27,000 JOD $\approx$ **38,000 USD** submitted to The Higher Council for Science & Technology (HCST), Jordan
- “Breaking Barriers: Ethical Solutions for Sexual & Reproductive Health (SRH) in Al Zaatari Camp's Syrian Female Refugees Community” (2024): **8,000 USD** submitted to Rawabit Partnership Development Grant- Rawabit U.S.-Jordan Higher Education Network (HEN).

Teaching Experience & Supervision

- **Taught** undergraduate and graduate courses including Medical Ethics & Regulatory Affairs, Human Biology, Anatomy and Physiology, Clinical Pathology, Clinical Immunology & Serology, Biotechnology & Bioprocess Engineering, Health Care Management, Bioorganic Chemistry, Pharmaceutical Microbiology, Biomaterials, Tissue Engineering, and Hazardous Waste Management. These courses were taught at the *German Jordanian University (GJU)*, *American University of Madaba (AUM)*, and *Al-Zytoonah University of Jordan (ZUJ)*.

Undergraduate Supervision (bachelor's degree)

- **Primary supervisor** for **40+ undergraduate (Bachelor's) graduation research projects** in biomedical sciences, some of them are in collaboration with local industry and/or Germany.
Selected Graduation Projects (with publications, features, or collaborations)
“Design and implementation of a Bioreactor.”
– Published & Featured by BBC Arabic
“Gut microbial community of phenylketonuria patients pilot study on Jordan children.”
– Published

- “Oral microbial community in the mouth of smokers.”
- Published
- “Mucus study during ventilation and anesthesia.”
- Project with Dräger-Germany
- “Dissolving of plastic bags made of polyethylene.”
- Conference poster in France
- “Production of biodiesel from the green marine macroalgae *Ulva lactuca*.”
- Published
- “Cleaning validation of Siemens Healthcare CT tables.”
- Project with Siemens-Germany

Last 3 years` Teaching Evaluation by students at GJU

Academic Year / Semester	Sections	Students Enrolled	Weighted Avg.
2022/2023 – Second	4	100	90.77%
2023/2024 – First	6	126	91.96%
2023/2024 – Second	5	139	93.34%
2024/2025 – First	2	44	94.44%
2024/2025 – Second	2	30	94.80%
2025/2026 – First (Interim)	4	73	96.86%
2025/2026 – First (Post)	4	73	92.57%
Grand Total	27	585	93.2%

M.Sc. Supervision & Examination (Inter-Institutional)

- **Primary supervisor for 1 M.Sc. thesis**
- **Co-supervisor for 3 M.Sc. theses**
- **External examiner for 2 M.Sc. theses**

These supervisions were conducted **inter-institutionally**, as there is **no** M.Sc. or P.hD. programs within my department.

Supervised & Cosupervised Theses & Dissertations

“The impact of locally designed magnifying reading glasses on reading speed in children with low vision.”

- Student: Duaa Yousef Al-Yousef
- Degree: M.Sc. in Vision Rehabilitation
- Year: 2019

“Investigation on the use of combined medications loading in polymeric films (Effect of drug load and polymer type).”

- Student: Aisha Kanaan
- Degree: M.Sc. in Pharmaceutical Sciences
- Year: 2023

“Exploring the Potential of Selected Long Non-coding RNAs as Diagnostic Biomarkers Among Chronic Lymphocytic Leukemia Patients and their Association with Vitamin D and Vitamin D Receptor Levels.”

- Student: Basil Samawi
- Degree: M.Sc. in Medical Laboratory Sciences
- Year: 2025

“Explainable AI-Based Decision-Making for Biomedical Quality Systems Using Data-Driven Models Aligned with ISO 9001:2015 Standard.”

- Student: Salma Alrawashdeh
- Degree: M.Sc. in Engineering Management
- Year: 2026

Professional, University and Community Service

Administrative Leadership

- **Acting Head**, Biomedical Engineering Department (2017/2018-2019/2020)
- **Acting Vice Dean**, School of Applied Medical Sciences (2018/2019-2019/2020)
- **Assistant Dean** for Accreditation and Quality Assurance (2017/2018-2018/2019)
- **Acting Dean and Chair**, School Council, School of Applied Medical Sciences, attained the German accreditation (ASIIN) for the two programs at the school, the Biomedical Engineering & Pharmaceutical and Chemical Engineering programs, in addition to getting the initial approvals for establishing a new M.Sc. program in Pharmaceutical & Chemical Engineering (2020/2021)
- **Chair**, Examinations Management Committee, Biomedical Engineering Department (2021)
- **Vice Chair**, Main Supplies Purchasing Committee (2024)
- **Secretary**, Biomedical Engineering Department Council (2024/2025)
- **Vice Chair**, Institutional Review Board (**IRB**), German Jordanian University (2021/2022)-JFDA approved
- **Chair**, Institutional Review Board (**IRB**) & **Assistant to the Dean of Scientific Research** for Research Ethics, German Jordanian University (from January 2025)-JFDA approved
- **Chair**, Main Supplies Purchasing Committee (from March 2025-March 2026)

University Governance and Committees

School & Department Councils

- Member/Representative, **School Council**, School of Applied Medical Sciences (2017/2018, 2020/2021, 2025/2026)
- Member, **University Council** (2017/2018, 2023/2024)
- Secretary, Biomedical Engineering **Department Council** (2024/2025)
- Research & Academic Committees
- Rapporteur, Department **Scientific Research** Committee (2019/2020, 2023/2024, 2024/2025)
- Member, Department **Appointment & Promotion** Committee (2023/2024)
- Member, Department **Graduation Projects & Activities** Committee (2023/2024)
- Member, Technical Committee for **Promotion Application Review** (2025)
- Member, Committee for Reviewing Scientific Leave (**Sabbatical**) Reports (2023, 2025)
- Member, Academic **Activities & Student Events Scheduling** Committee (2025/2026)

Quality Assurance & Accreditation

- Rapporteur, **ASIIN German Accreditation** Report Review Committee (2021)
- Rapporteur, Department **National Qualifications Framework** Alignment Committee (2023/2024)
- Rapporteur, **Internal Audit** Committees (ISO 9001:2015 & ISO 21001:2018) (2024–2025)

Strategic & University-Wide Initiatives

- Member, University **Risk Management** Committee (2024)
- Member, Working Team for **King Abdullah II Award for Excellence** - Risk Management (2025)
- Member, **USAID Higher Education Project Application** Committee (2024)

Additional Committee Roles

- Member, **Dual Studies** Committee (2021)
- Member, **Public Safety** Committee (2022-2023)
- Member, **School Safety** Committees, School of Applied Medical Sciences (2024-2025)
- Member, **Class Transfer** Committee (2024)
- Member, **Chemical Analyzer** Committee (2022)

Activities Directed to or from the National Community

- **WHO COVID-19 Database Contribution** - Provided biomedical and epidemiological data for global pandemic tracking. World Health Organization, 2023.
- **Authorship of Science Articles in *Al-Rai Newspaper*** - Published articles on COVID-19 misinformation and prion disease for public education. Al-Rai Daily Newspaper, 2025.
- **Feature in *Times Higher Education (JOCOV-19)*** - Contributed to an article highlighting the JOCOV-19 viral RNA extraction kit. *Times Higher Education*, 2022.
- **Guest Speaker at JPSA Targeted Cancer Therapy Talk** - Delivered a presentation on targeted therapy approaches. JPSA, 2021.
- **Monkeypox Talk 2022** - Educated the community on the disease and public health measures. GJU, 2022.
- **Phi Science COVID Chat** - Discussed scientific aspects of COVID-19. Phi, 2021.
- **JoYS Training for Students** - Conducted training on scientific writing and research skills for school students. Jordan Young Scientist (JoYS), 2024.
- **KHCC HRPP Certification** - Completed Human Research Protection Program training. King Hussein Cancer Center (KHCC), 2024.
- **GeneTech CPD Certification** - Completed continuing professional development in genomics and biotechnology. GeneTech, 2024.
- **MTLS Transplant Training** - Gained specialized knowledge in transplant medicine. Medical Technology & Laboratory Society (MTLS), 2024.
- **NSC Workshop Participation** - Attended a workshop organized by the National Scientific Council. NSC, 2024.
- **NCRD Joint Committee Meeting** - Participated in a joint committee meeting with the National Center for Research and Development. NCRD, 2024.

- **JFDA Biologicals Committee** - Appointed to the Jordan Food and Drug Administration committee for biological products. JFDA, 2025.
- **External Consultancy at Jordanian Chinese Medicine Center** - Received university approval for external consultancy (10% revenue to university). GJU / Consultation and Training Center, 2022.
- **External Teaching at American University of Madaba** - Approved to teach Clinical Immunology & Clinical Pathology (6 credit hours). AUM / GJU President, 2022 & 2023.
- **Inclusion in Health Informatics Program Cadre** - Named in the faculty cadre for the new Health Informatics Bachelor program. School of Applied Medical Sciences, 2022.

Activities Directed to or from the International Community

- **Erasmus+ Greece Mobility Program** - Completed academic exchange, teaching, and research collaboration at Democritus University of Thrace. Erasmus+ / GJU, 2025.
- **Erasmus+ Selection Committee 2025** - Served on the committee evaluating applications for Erasmus+ staff mobility. Erasmus+ / GJU, 2025.
- **UiTM MoA Approval** - Contributed to the memorandum of agreement with Universiti Teknologi MARA (Malaysia). UiTM, 2025.
- **IREX HEIG Selection** - Part of the team selected for the USAID-funded Higher Education for Innovation and Growth project. IREX, 2025.
- **DAAD Info Session Guest Speaker** - Attended a session on DAAD opportunities to colleagues and students. DAAD, 2025.
- **DAAD Grant Proposal Writing Workshop** - Attended workshop on advanced techniques for research grant applications. DAAD, 2025.
- **DAAD Enterprising Teaching Workshop** - Participated in a workshop on innovative and entrepreneurial teaching methods. DAAD, 2025.
- **IEEE Tunisia Panel Discussion** - Served as a panelist on humanitarian technologies in a cross-sectional discussion. IEEE Tunisia Section, 2025.
- **IEEE WIE Workshop & Summit** - Participated and contributed to sessions on women in engineering leadership. IEEE Women in Engineering, 2025.
- **IEEE Authorship Symposium** - Attended symposium on responsible scientific publishing. IEEE, 2023.
- **UCSD Bioethics Workshop** - Participated in a workshop on bioethical issues in medical and research contexts. University of California San Diego (UCSD), 2020.
- **Ethics MENA Conference 2024** - Attended the conference on research ethics in the Middle East and North Africa. MENA Ethics Conference, 2024.
- **AI Ethics Workshop** - Explored ethical implications of artificial intelligence in biomedical and healthcare contexts. External organizers, 2025.
- **Peer Review for Cell Press Journals** - Provided expert manuscript evaluation for biomedical journals. Cell Press, 2022.
- **ACS Community Highlights** - Featured in the American Chemical Society journal for contributions to laboratory safety research. ACS, 2022.

Non-Academic Activities

- **Chair, IEEE SIGHT Jordan Section** - Led humanitarian technology projects, workshops, and digital-identity initiatives across Jordan, e.g., STAR Workshop and TECH 4 HUMAN SAKE, 2025.
- **Judge, IEEE EMBS Student Design Competition** - Evaluated biomedical engineering student projects and provided feedback. IEEE EMBS Jordan Chapter, 2024.
- **Founder & Advisor, Biomedical Engineering Student Club** - Provided academic guidance, event support, and professional development activities, 2020-2025.
- **Participation in GJU Open House Campaigns** - Represented the School of Applied Medical Sciences in multiple recruitment campaigns (2022, 2023/2024, 2024/2025, 2025/2026).
- **Hosted a Guest Lecture at Medical Hub** - Hosted a lecture on biomedical engineering topics. Medical Hub, 2022.
- **Hosted a Guest Lecture at VR Seminar (BME Club)** - Presented on virtual reality applications in biomedical engineering. BME Club / GJU, 2022.
- **Hosted a Guest Speaker at Qiagen Event** - Presented on biomedical research tools and applications. Qiagen, 2022.
- **Flying Faculty Hosting** - Hosted international (German) faculty for collaborative teaching. GJU, 2024.

Selected Awards

- **IEEE HTB Challenge 2025** - Participant in the Humanitarian Technology Board challenge. IEEE Humanitarian Technology Board, 2025.
- **GIMI Innovation Associate**, certified by Global Innovation Management Institute , 2025
- **The BioScience Excellence Awards** in Biotechnology, 4th edition, 2024
- **Certificate of Appreciation** for SDGs-related Research, Elsevier , 2023
- **Patent Application** JO|P|2020|171 - JFDA-approved Viral RNA Extraction Kit (JOCOV-19); featured by [Nature](#) Middle East & [THE](#) (Times Higher Education), 2020
- **Funded One-Year Research Fellowship**, University of California San Diego (UCSD), USA, 2018
- **Paper of the Month**, University of New South Wales , 2015
- **Postgraduate Research Student Support (PRSS) Scheme** funding, UNSW- Round 1, 2014
- **School of Medical Sciences Travel Award**, University of New South Wales, 2013
- **Tuition Fees Scholarship (TFS)** for international students to study for a PhD, from the University of New South Wales, Australia, 2011
- **GJU President Appreciation Letter** - For service and leadership contributions. GJU President, 2022.
- **eHDA (eHealth Development Association) Volunteer Appreciation Certificate** - For volunteer service with the eHealth Development Association. eHDA, 2022.
- **DAAD PhD Selection Committee Membership** - Appointed to evaluate and select candidates for DAAD doctoral scholarships. DAAD, 2023 & 2024.
- **GJU Recognition Ceremony** - Invited participant for outstanding service and achievements. GJU President, 2025.

- **IEEE Senior Member** - Official elevation to Senior Member status for significant contributions to biomedical engineering. IEEE, 2025.

Professional Development

- Earned certifications in **Python** and **R** Programming - DataCamp, 2024
- Completed the **Entrepreneurial Teaching Training of Trainers (ToT)** Workshop - DAAD Germany, 2019
- Achieved **Certificate IV in Training and Assessment (TAE40110)** - Australia, 2014
- Completed **Foundations of University Learning and Teaching (FULT)** - UNSW Australia, 2013
- Completed a 4-month **Bioinformatics Specialization program** - Phi Science, 2022
- Completed a 3-day **Genomics Data Science** Training - Helix Biogen Institute, 2025
- Completed the **Digital Teaching and Blended Learning program** - GIZ / Steinbeis Transfer Center, 2020
- Completed **Digital Learning & Digital Transformation** training - CeLAPI, GJU, 2024-2025
- Participated in the **Bioinformatics Research Ethics Workshop** - UCSD / Fogarty NIH, 2020
- Participated in the **Improving the Peer Review Experience** workshop - Cell Press, 2024
- Participated in the **CPD/CME COVID-19 Vaccines** webinar - WSPID / EACCME, 2020
- Participated in the **Managing the Ethics of AI in Organisations** webinar - Good Governance Academy, 2025
- Completed **Biosimilar Technical & Regulatory Training** - ZoneTechs, 2025
- Participated in the **Entrepreneurship Bootcamp** - DI-TECH (GJU), 2023
- Participated in **IRB Workshop** and **DSR Workshop on Research Ethics** - GJU / DSR, 2024 & 2025

Professional memberships

- **IEEE**, Senior Member, SIGHT Chair & Member, EMBS Member, Jordan Section Coordinator
- **Australian Society for Biochemistry & Molecular Biology**
- **E-Health Development Association**, Jordan
- Membership of The **International Advisory Board** for Jordan Journal of Biological Sciences (JJBS) affiliated to the Higher Education Council and hosted by the Hashemite University, Jordan
- **Jordanian Society of Genetic Engineers**, Jordan
- Fellow, **The Arab Academy for Responsible Conduct of Research**

Publications

First-Author Publications (Chronological Order, Newest to Oldest)

- **Al-Zyoud W**, Abu Ismail H, Althawbih H, Odeh B, Saket M, et al. Insights for a next pandemic: An assessment of laboratory COVID-19 safety. *Safety*. 11(1):13. **2025**.
- **Al-Zyoud W**, Hamzeh S, Masri D, Saket M, Qader JA, Al-Qaraghuli R, et al. Exploring the potential of *Ulva lactuca* from the Gulf of Aqaba for micro-scale biodiesel production:

Designing a small-scale photo-bioreactor. *Jordan Journal of Biological Sciences (JJBS)*. 17(2). 2024.

- **Al-Zyoud W**, Haddadin D, Hasan SA, Jaradat H, Kanoun O. Biocompatibility testing for implants: A novel tool for selection and characterization. *Materials*. 16(21):6881. 2023.
- **Al-Zyoud W**. The psychological effects of physicians' communication skills on COVID-19 patients [Response to Letter]. *Patient Preference and Adherence*. 1729-1730. 2023.
- **Al-Zyoud W**, Erekat D, Saraiji R. COVID-19 chest X-ray image analysis by threshold-based segmentation. *Heliyon*. 9(3). 2023.
- **Al-Zyoud W**, Haddad H. Potential linear B-cells epitope change to a helix structure in the spike of Omicron 21L or BA.2 predicts increased SARS-CoV-2 antibodies evasion. *Virology*. 573:84-95. 2022.
- **Al-Zyoud W**, Oweis T, Al-Thawabih H, Al-Saqqar F, Al-Kazwini A, et al. The psychological effects of physicians' communication skills on COVID-19 patients. *Patient Preference and Adherence*. 677-690. 2021.
- **Al-Zyoud W**, Haddad H. Dynamics prediction of emerging notable spike protein mutations in SARS-CoV-2 implies a need for updated vaccines. *Biochimie*. 191:91-103. 2021.
- **Al-Zyoud W**, Haddad H. Mutational sensitivity of D614G in spike protein of SARS-CoV-2 in Jordan. *Biochemistry and Biophysics Reports*. 25:100896. 2021.
- **Al-Zyoud W**, Hajjo R, Abu-Siniyeh A, Hajjaj S. Salivary microbiome and cigarette smoking: a first of its kind investigation in Jordan. *International Journal of Environmental Research & Public Health*. 17(1):256. 2020.
- **Al-Zyoud W**, Helou AA, AlQasem E, Rawashdeh NA. Visual feature extraction from dermoscopic colour images for classification of melanocytic skin lesions. *EurAsian Journal of BioSciences*. 14:1299. 2020.
- **Al-Zyoud W**, Qunies AM, Walters AUC, Jalsa NK. Perceptions of chemical safety in laboratories. *Safety*. 5(2):21. 2019.
- **Al-Zyoud W**, Nasereddin A, Aljarajrah H, Saket M. Culturable gut bacteria lack Escherichia coli in children with phenylketonuria. *New Microbes and New Infections*. 32:100616. 2019.
- **Al-Zyoud W**, Mansour S, Elayan F, Homsy N, Ajeilat H, Dajani R. Conflict of interest in research at universities: a pioneer investigation in Jordan. *International Journal of Education and Research*. 3(9):1-11. 2019.
- **Al-Zyoud WA**, Hynson RMG, Ganuelas LA, Coster ACF, Duff AP, et al. Binding of transcription factor GabR to DNA requires recognition of DNA shape at a location distinct from its cognate binding site. *Nucleic Acids Research*. 44(3):1411-1420. 2016.
- **Al-Zyoud W**, Hynson R, Duff A, Sobey T, Lee L, Liu D, Böcking T. Small-angle X-ray scattering study for conformational changes of unusual transcription factor at the operator. *Biophysical Journal*. 106(2):497a. 2014.

Co-Author Publications (Second Author or Later, Chronological Order, Newest to Oldest)

- Al-Ameer HJ, Zihlif M, Maslat A, Al-Awaida WJ, Ayyash AM, Imraish A, **Al-Zyoud W**, et al. Targeting the proliferation of glioblastoma cells and enhancement of doxorubicin and temozolomide cytotoxicity through inhibition of PFKFB4 and HMOX1 genes with siRNAs. *Scientific Reports*. 15(1):27861. **2025**.
- Haddad H, **Al-Zyoud W**. Prion propensity of Betacoronaviruses including SARS-CoV-2. *Heliyon*. 11: e42199. **2025**.
- Abu-Siniyeh A, Khataibeh M, **Al-Zyoud W**, Al Holi M. Zebrafish as a model for human epithelial pathology. *Laboratory Animal Research*. 41(1):6. **2025**.
- Tarawneh O, Kanan A, **Al-Zyoud W**, Khanfar M, Hailat M, Alhusban AA, et al. Innovative buccal film delivery system for hypertension: Combining candesartan and hydrochlorothiazide. *Journal of Biomaterials Applications*. 39(9):1014-1029. **2025**.
- Abu-Safe H, **Al-Zyoud W**, Al-Adamat K, Haddad A, Al-Sabbagh M, Saleh A, et al. Identifying human ABO blood type using z-scan technique. *Measurement*. 240:115571. **2024**.
- Bekawi D, Gierke AM, Hasan SA, **Al-Zyoud W**, Hessling M. Photoinactivation of *Streptococcus cristatus* and *Streptococcus vestibularis* by 222 and 254 nm UVC irradiation. *Acta Microbiologica Bulgarica*. 40(4):563-567. **2024**.
- Al-Kazwini AT, **Al-Zyoud W**, Al-Nasser S, Alswaiti SA, Khankan DS, et al. How did Jordan combat COVID-19: Lessons learned for scaling. *Med*. 2(4):348-349. **2021**.
- Haddad H, **Al-Zyoud W**. miRNA target prediction might explain the reduced transmission of SARS-CoV-2 in Jordan, Middle East. *Non-coding RNA Research*. 5(3):135-143. **2020**.
- Abu-Siniyeh A, **Al-Zyoud W**. Highlights on selected microscopy techniques to study zebrafish developmental biology. *Laboratory Animal Research*. 36(1):12. **2020**.
- Qananwah Q, **Al-Zyoud W**, Al-Zaben A. Biomedical invasive pressure sensor coatings: calibration and waveform perspectives. *Journal of Medical Engineering & Technology*. 44(4):203-209. **2020**.
- Al-Mashaqbah SF, **Al-Zyoud W**, Al-Zaben A. Utilizing upper arm images in determining body mass index and nutritional status. *Engineering, Technology & Applied Science Research*. 9:4392-4396. **2019**.
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Global Burden of Disease Collaborations (Chronological Order, Newest to Oldest)

Note: As a contributing author to multiple Global Burden of Disease (GBD) studies published in *The Lancet*, *The Lancet Infectious Diseases*, *The Lancet Respiratory Medicine*, and *Nature Medicine*, I provided subject matter expertise and regional data interpretation for Jordan and the Middle East, helping to shape global health metrics estimates of disease burden, antimicrobial resistance, and risk factors in this region.

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Conference Publications, Presentations & Organizations

- **Al-Zyoud W**, Al-Awamleh J, Asalwa B, Bater R. Smoking cessation by virtual reality-a VR integrated app concept. *IEEE 2nd National Biomedical Engineering Conference (NBEC)*, Melaka, Malaysia, **2023**.
- **Al-Zyoud W**, Bater R, Awad T, Yasin O, Al-Zaben A. Bioreactor for biopharmaceutical production: Simple controlled environment design, *2018 IEEE 4th Middle East Conference on Biomedical Engineering (MECBME)*, Tunis, Tunisia, **2018**.
- Al-Kazwini A, Halilah A, Hendi A, Al-Saeed I, Al-Azamat A, **Al-Zyoud W**. Frequency detection of hand tremors using customized accelerometer, *IEEE National Biomedical Engineering Conference (NBEC)*, Kuala Lumpur, Malaysia, **2021**.
- Albarahmieh E, Alziadat R, Saket M, Khanfar M, **Al-Zyoud W**. Streamline development of propellant-free topical pharmaceutical foam via quality by design approach for potential sunscreen application. *Proceedings of the 1st International Congress on Engineering Technologies by Taylor & Francis, JUST, Jordan*, **2021**.
- **Poster presentation**, *ECDS (Emerging Concepts & Design Sustainability) Conference*, 2023, Villers-sur-Mer, Calvados, France, Poster #P4
- **Abstract/Poster presentation**, 58th *Biophysical Society Meeting*, **2014**, San Francisco, USA, Poster #2516
- **Poster presentation**, *ComBio 2012*, Adelaide Convention Centre, Adelaide, South Australia, Poster #POS-MON-016, **2012**
- **Abstract**, the 1st international scientific conference on "The Red Sea *Ecosphere*: Conservation and Management of Red Sea Marine Environment," in the city of Aqaba, **2022**
- **Orgnaizing & Scientific committee member** for the 5th IEEE Middle East and Africa Conference on Biomedical Engineering (*MECBME*), Yarmouk University, IEEE JS & IEEE JS EMBS, Jordan, **2020**

- **Technical committee member** for the 10th, 11th, 12th, 13th & 14th International Conference on Biomedical Engineering and Technology, IEEE, ([ICBET 2020, 2021, 2022 & 2023](#), Tokyo, Japan) & ([ICBET 2024](#), Seoul, South Korea), reviewing conference submissions that were published by The Association for Computing Machinery, NY, USA

Book

- **Al-Zyoud**, DNA binding mechanism of an unusual bacterial transcription regulator, [Lambert Academic publishing](#), ISBN 13-9783659819513, **2016**

Book Chapter

- **Al-Zyoud W**, Erekat D. Dermatoscopy and other practices. *Dermatoscopy*. [IntechOpen](#) 91. **2022**

Media Exposure

- **Newspaper Articles** | [Alrai](#) (Jordan) | 2021–2024
Wrote and presented articles on public health, pandemic awareness and COVID-19.
- **Interview** | [TAJ NEWS](#), [ADDUSTOUR](#), [ALGHAD](#) (Jordan) | 2021–2024
Discussed pandemic awareness, public health, and COVID-19 related topics.
- **Television Interview** | [Alhurra TV](#) | 2024
Interview featured on LinkedIn (topic: antimicrobial resistance / biomedical research).
- **Media Feature** | [BBC Arabic](#) | 2018
Featured for graduation project “Design and implementation of a Bioreactor.”
- **Research Highlight** | [Nature Middle East](#) & [Alfanar Media](#) | 2020
Featured for JFDA-approved viral RNA extraction kit (JOCOV-19).
- **Media Mention** | *The Times* | 2020
Covered the JOCOV-19 viral RNA extraction kit.
- **Media Feature** | [The Guardian](#) | 2024
Cited as a contributing researcher in the global antimicrobial resistance (AMR) study published in *The Lancet*, forecasting superbug deaths and highlighting global health trends.
- **Presentation to Royalty** | His [Royal](#) Highness Prince Al-Hasan bin Talal | 2023
Presented research on biodiesel production from algae of Aqaba
- Presented a graduation project to His [Majesty](#) King Abdullah II | 2025